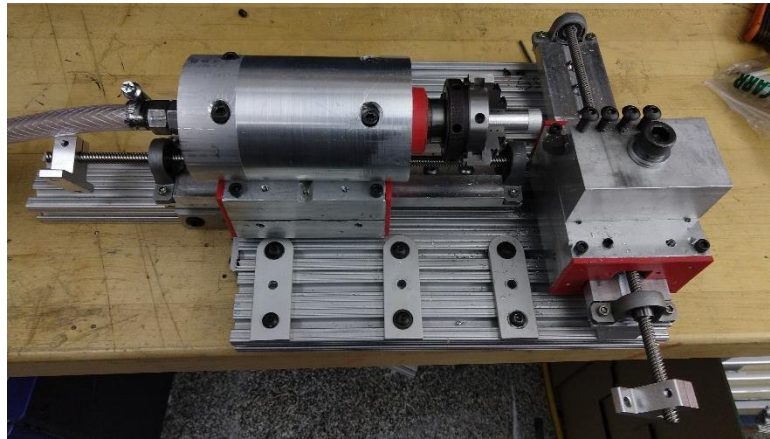


1. Executive Summary

My objective this semester as a part of the Precision Machine Design course, taught by Professor Alexander Slocum, was to model, design and build a T-Based lathe. As part of this objective, there were many learning opportunities and challenges presented that allowed me to build skills in developing error budgets, error analysis using homogeneous transformation matrices, and designing and integrating mechanical components and predicting their behavior.



1.a Overview of a T-Based Lathe

In a traditional lathe, the spindle holding the part to be cut is fixed in place and the tool moves along two axes of movement. However, in a T-based lathe, the spindle is mounted to one axis of movement and the tool upon a different, axis that is perpendicular to the spindle, creating a 'T' shape.

1.b The Linear Motion Modules

I chose to pursue a boxway linear motion slide for my linear motion modules. The initial results from the testing of my sketch models led me to conclude that a boxway would provide me the best combination of stiffness, ease of manufacturing and modeling simplicity. The carriages are made out of two plates of aluminum, each approximately 1" thick. I chose to use Delrin as bearing pads based on its stiffness as well as low coefficient of friction. Set screws are used to preload the carriage against the rails and prevent rattling. A cutout is made along the bottom of the carriage to provide a space for a leadscrew to be attached to the carriage.

1.c The Spindle

During the class, we had a visit from some representatives from Igus (www.igus.com) and I wanted to try to J3 plain bearings they gave us into the design of my spindle. I turned the spindle housing out of 3" diameter stock 6061-T6 aluminum and milled a flat face in order to mount the spindle housing to one of the linear motion module carriages. The spindle shaft was fabricated with M14x1 external threads so that I could attach a mini 3-jaw lathe chuck to hold the part. The spindle shaft is preloaded using two

disk spring washers as well as two nuts acting together to form a lock-nut. I decided to pursue testing the lathe by cutting parts in the jaws of the chuck as well as threaded directly onto the spindle shaft.

1.d The Base Plate

I decided to use 80/20 extrusion as a base due to the high stiffness compared to wood and the lack of availability of a sufficiently large piece of stock material. By assembling the 80/20 such that the cutting forces act 'with the grain' of the extrusion slot, the separate pieces of 80/20 act as a single base plate.

1.e The Tool Holder

The tool holder was modeled after the tool holders found on standard machine shop lathes. Four $\frac{1}{4}$ -20 screws provide the clamping force needed to hold the tool in place during the cutting process. A $\frac{1}{2}$ -13 bolt hole threaded into the carriage allows the tool holder to be clamped to the carriage as well as allow for angle adjustment to the tool.

1.f Actuation of the Linear Motion Modules via Leadscrew

The lead screws were mounted using the pillow blocks that came with the set (\$13 on Amazon). Although this is not a 'proper' mounting technique, it is sufficient for my application. If my lathe were to be sent to mass production, I would develop proper bearing supports and support blocks. The model for the stiffness of my lead screw is based off of the work done by my classmates, Akshay Harlalka, for his rotary motion module.

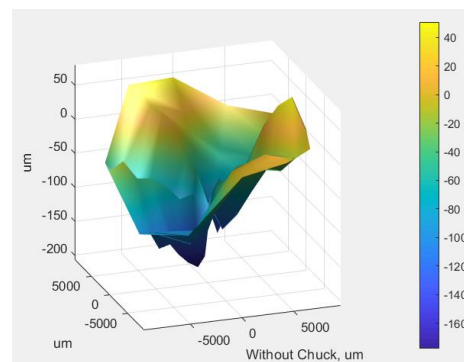
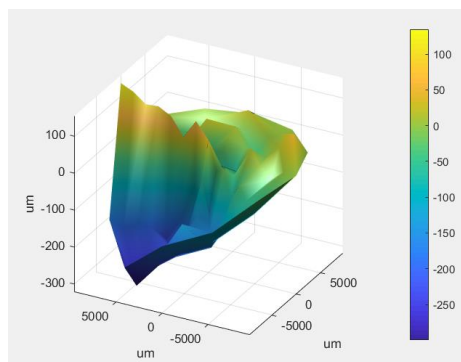
1.g Error Model of the T-Based Lathe

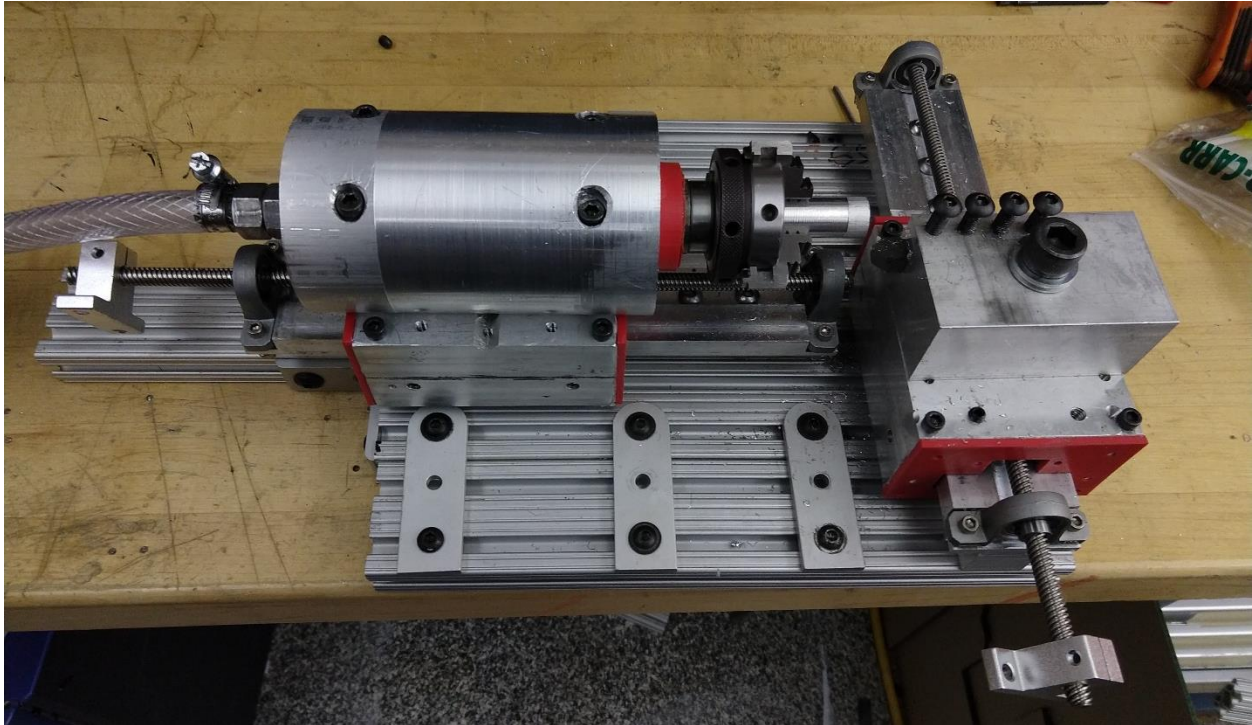
The error motions due to rail parallelism and applied loads were estimated for the linear motion module as well as the spindle. The error motions of each coordinate system can be transformed into error motions at the part/tool interface using homogenous transformation matrices.

1.g Testing of the T-Based Lathe

The cutting forces calculated for facing an Aluminum part 1" in diameter with a pass of 1 mm were calculated to be 100 N tangential cutting, and 70 N for the radial and thrust forces respectively using the equations outlined by Kalpakjian [1].

Facing passes were taken on parts, both with and without the use of the three-jaw chuck. The part faced with the three-jaw chuck was visibly more 'wavy' than the part threaded directly onto the spindle shaft.





2. Expected Cutting Forces

Using the analysis described in Manufacturing Engineering and Technology by Kalpakjian and Schmid [1], I created a spreadsheet to predict the cutting forces that my linear motion slide would experience.

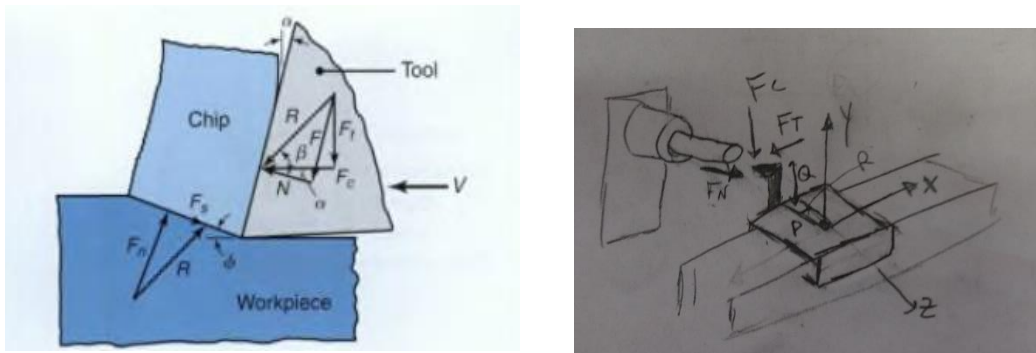


Figure 1. The cutting forces and geometry experienced in the facing process of a T-based lathe

This calculation is described in [1] in chapters 21 and 23 and the full analysis can be seen in my analysis spreadsheet.

Table 1. Summary of cutting forces analysis for facing an aluminum part

Material	Aluminum
Spindle Speed	200 RPM
Depth of Cut	1 mm

Feed rate	.33 mm/s
Cutting Force (F_y) (Predicted)	100 N
Thrust Force (F_x) (Predicted)	70 N
Normal Force (F_z) (Predicted)	70 N

Based on these predicted cutting forces, it was necessary to then use the axis error apportionment to calculate the allowable bearing and structure deflections due to loads. For my lathe to have 100 microns of total allowable error, the total load induced deflection due to the bearings and structure for one of my axis is 19.9 μm total. Therefore, using the expected forces and the allowable deflection, an initial stiffness requirement can be calculated. Using the resultant force seen in the cutting process, the net system stiffness of the system is calculated to be 6 N/ μm , which is a reasonable goal.

3. Design of the Linear Motion Module

I began my design of the linear motion module (LMM) by developing a FRDPARRC as well as testing several sketch model concepts.

Functional Requirement	Design Parameter	Analysis	References	Risks	Counter Measures
Stiffness of 9000 N-mm along direction of cutting forces	> Guide Rail parameters (length, width, cross section, material)	> Beam Bending and Deflections	> Roark's Formulas for Stress and Strain	> I miss a key element in the prediction and design of the linear	> Be very vigorous in analysis and using HTMs
	> Bearing Parameters (Stiffness, Configuration, Type)	> FEA of carriage	> http://www.solidworks.com/sw/product/simulation/finite-element-analysis.htm	> I am unable to machine the LMS parts to the required tolerances	> Use CNCs whenever possible and keep design system to have loose tolerance
	> Carriage parameters	> Sensitive Directions and Structural Loop	> PMD		
Range of Travel between 100-200 mm	> Stiffness and Cost	> Beam Bending and Deflections	> Roark's Formulas for Stress and Strain	> Certain designs have varying stiffness depending on carriage location	> Either design a constant stiffness LMS or design cutting range of travel to be at a certain point
Cost less than \$50	Ratio of 'Goodness' to \$\$\$	Cost analysis of rolling vs sliding + material choices and availability of free parts	> McMaster-Carr, Igus, SDP/SI, Misumi	> I am unable to design a proper lathe because I don't want to spend big bucks	> Shape design around easily findable parts and stock in lab
Driven By a leadscrew + Stepper Motor	> Carriage Configuration	> Screw Force Calculation	> ScrewForce.xlsx	> Cutting forces are stronger than predicted, carriage can't move	> Thorough analysis of loads expected to be seen by carriage and frictional forces
	> System Friction > Cutting Forces	> Lathe Turning Force Calculation	> Roark, > Machinery's Handbook		
Running Parallelism Better than 0.1 Deg	> Tolerance of manufacturing rails	> Forces on bearings due to parallelism of rails	> PMD	> I struggle to mount the rails perfectly parallel and the system binds at the ends of the travel length	> Utilize EACs to ensure precise alignment
	> Rail Alignment during installation	> EAC design for rail mounting			

Figure 2 FRDPARRC for the Linear Motion Module

	Concept 1	Concept 2	Concept 3
	U-Groove Bearings	2 Rail Slider	T-Rail
Angular Straightness			
Yaw (rad, deg)	0.003 0.175	0.002 0.131	0.002 0.116
Pitch (rad, deg)	0.001 0.057	0.003 0.189	0.001 0.057
Stiffness			
Y-Dir (Nm)	23113.9	65616.8	NA
Z-Dir (Nm)	1036054.7	26246.7	NA
Pitch (Nrad)	1533.4	67.7	NA



Stiffness testing using a dial indicator and applied weight and straightness testing using a laser pointer

Figure3 Concept sketch comparison of stiffness and straightness

The main conclusion that I drew from testing the sketch models as that the two-rail slider was the most sensitive to manufacturing errors. Furthermore, the stiffness of the rail sliding module varies depending on the position of the carriage on the rail. I therefore chose to pursue a boxway design.

3.1. Design of the Carriage and Rail of the LMM

The LMM consists of two main components, the carriage and the T-beam. The carriage is made out of pieces of 25.4mm x 87mm x 101mm aluminum bolted and JB Welded together during the manufacturing process. The T-rail consists of a rectangular piece of aluminum stock with a milled slot down the underside. A square piece of aluminum stock is lightly press fit and JB Welded into this slot, forming the 'T'. After the epoxy cured, bolt holes were drilled through both pieces to bolt them together and to the base plate.

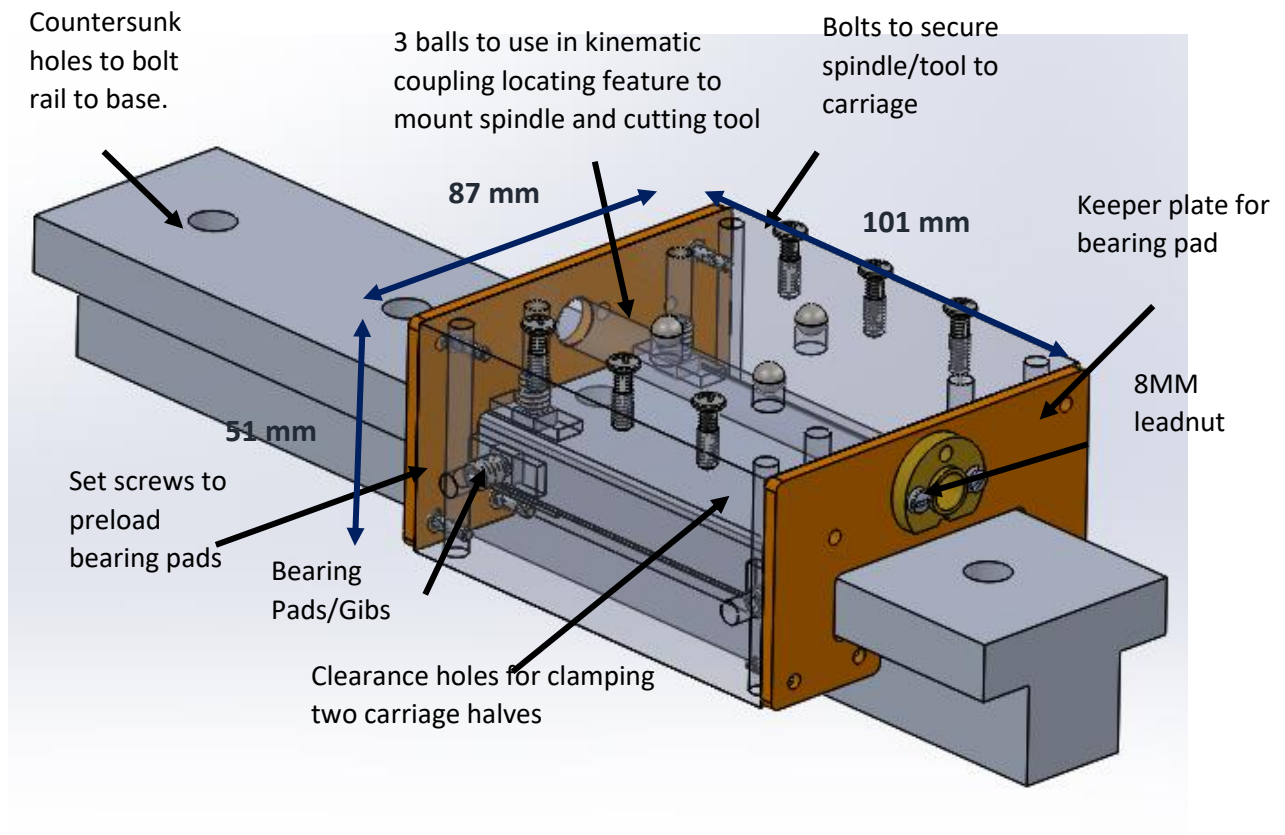


Figure 4. Linear Motion Slide Original Design

3.2. Manufacturing the LMM

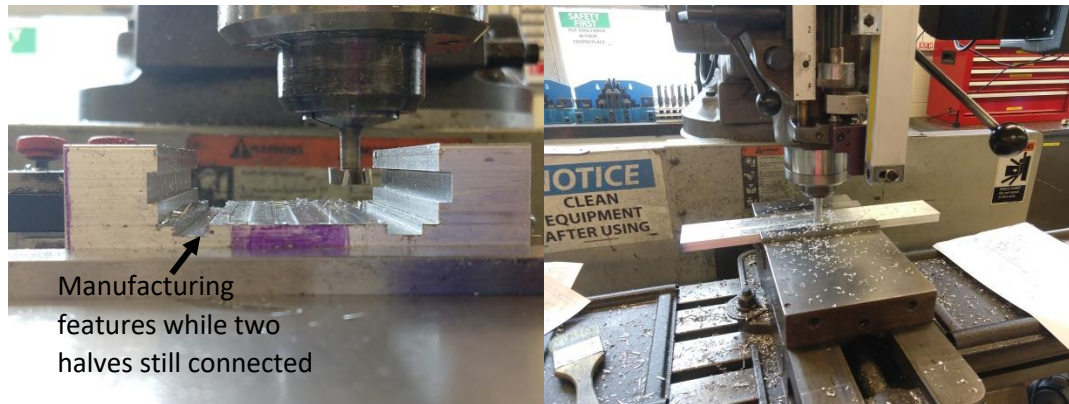


Figure 5. Manufacturing the slots for the gibs and bearing pads (left) and manufacturing an insert for the T-rail (right)

The two most important aspects of the manufacturing process that I would like to elaborate on are the addition of features to test the stiffness and accuracy of the LMS and the manufacturing of the side and bottom plates of the LMS carriage.

I knew going in to the machining process that it would be necessary to include features that would allow me to test the stiffness and accuracy of my LMS. To test the stiffness of the carriage rotationally and translationally, I knew I would need to apply a moment and a force at different points on the carriage. I determined that all stiffness values could be determined by including a $\frac{1}{2}$ -13 tapped bolt hole on the top and side of the carriage. Using an eye-hook bolt, I determined that by pulling on the two different bolt configurations in different directions I could apply a specific, known, moment or force load about a particular axis. I also incorporated perpendicular V-grooves to securely mount a laser pointer along the x and z axes of the carriage to measure rotational stiffness as well as linear yaw and pitch accuracy and repeatability.

The second manufacturing technique that I found particularly helpful was machining the side and keeper plates of the carriage out of a single piece of aluminum stock. By machining the pieces this way, I could machine all of the features (slots for the bearing pads as well as the slots for the T-rail) at once and then I could separate the two halves quite accurately and with assurance of symmetry of the parts. This technique can be seen in Fig. 2.

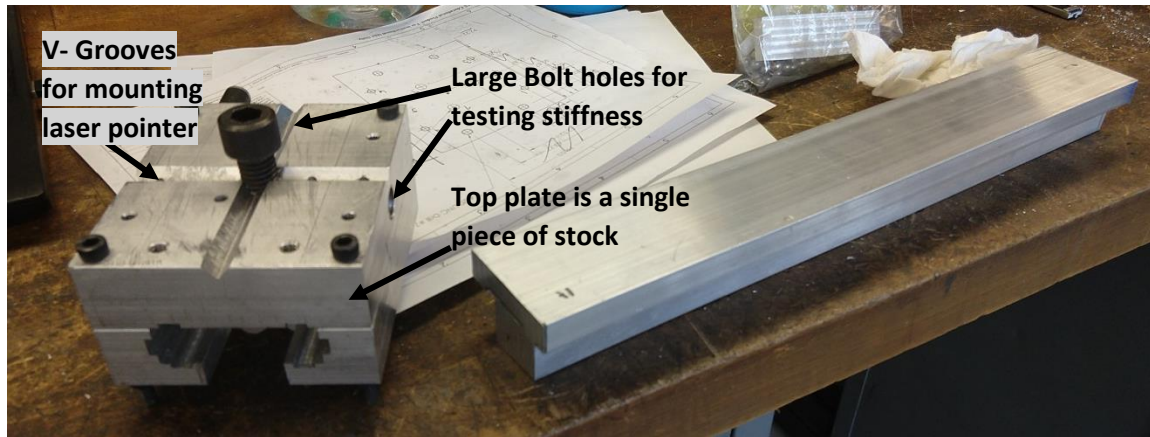


Figure 6. Finished LMS assembly

3.3 Testing the LMM

Using 100N spring scales, the rotational stiffness as well as the linear stiffness of the carriage was tested. The data and calculations for these measurements can be seen in LatheDesign_RMH.xlsx in the sheet "LMSTesting".

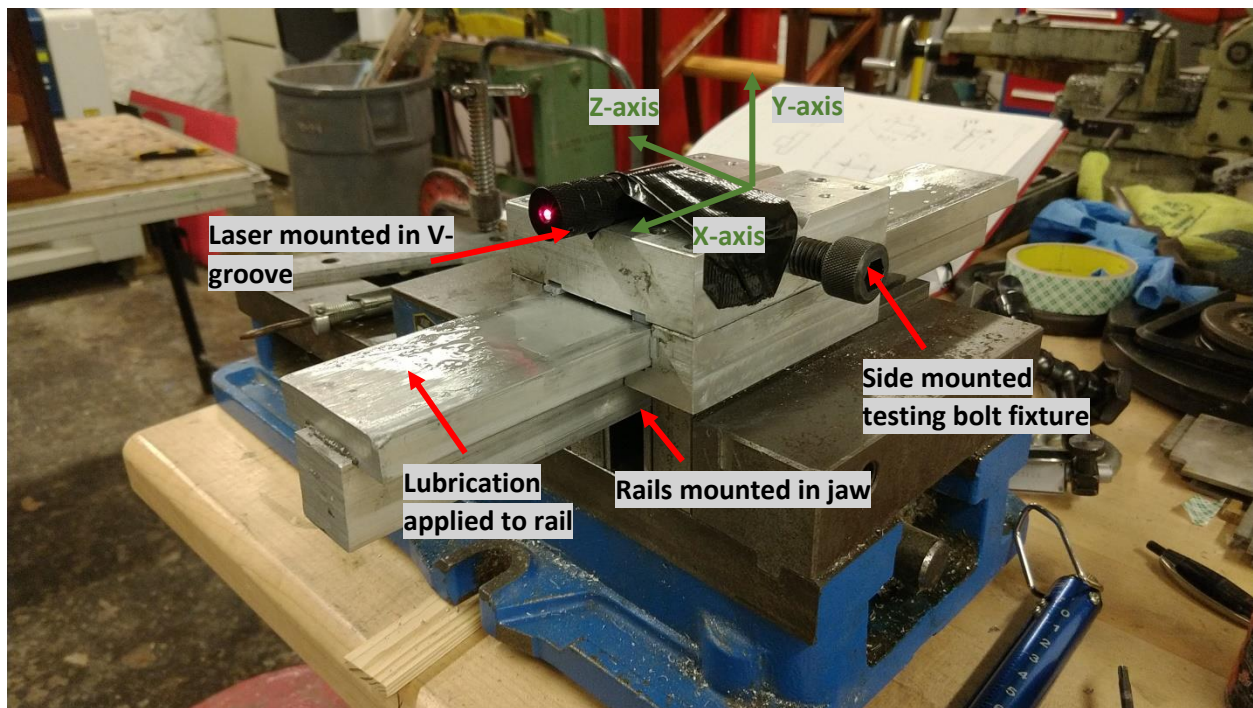


Figure 7. Testing setup for LMS module

The rotational stiffness was tested by applying a moment about the x,y and z-axes separately and measuring the deflection using a laser pointer projected 8 meters away.

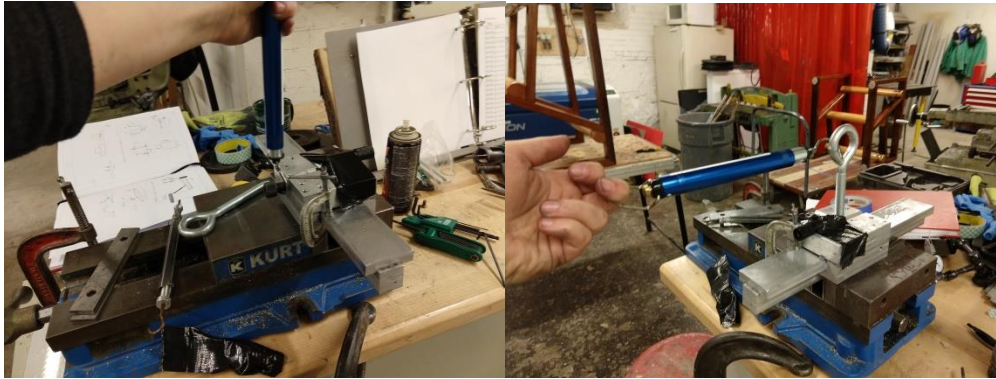


Figure 8. Testing Roll and pitch stiffness of LMS

The linear stiffness along the Y and Z axes was found by using a dial indicator fixed to the base of the vice and measuring the deflection as a linear force was applied along the axes of interest. However, using the tools available, I could not notice a measurable deflection in either the Y or Z axes, meaning that my system is much stiffer than what I can measure.

The pitch (Θ_Z) and yaw (Θ_Y) accuracy and repeatability were found by marking the locations of the laser pointer as the carriage traveled along the length of the rail. A point was marked when the carriage was at both of its extremes in travel distance along the rail.

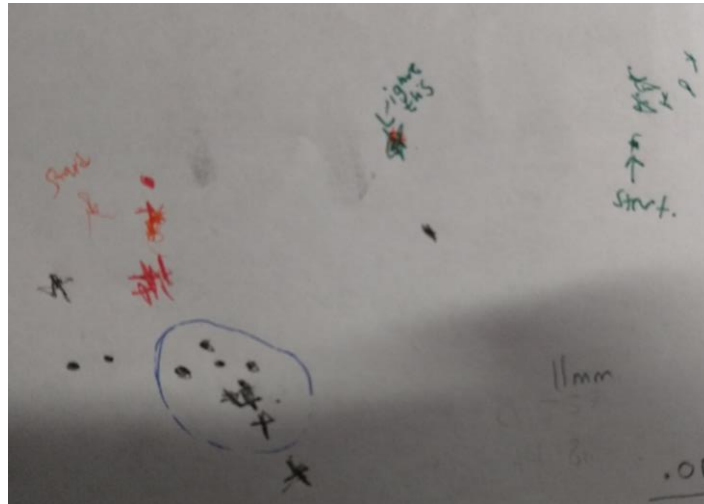


Figure 9. Laser pointer data collection for angular stiffness, accuracy and repeatability

Table 2. Summary of Stiffness testing data (LMS_Analysis_RMH.xlsx)

	Theta Y (Yaw) (N rad/m)	Theta Z (Pitch) (N rad/m)	Theta X (Roll) (N rad/m)	Linear Y Stiffness (N /m)	Linear Z stiffness (N /m)
Calculated Stiffness	1.20E+05	1.92E+04	3.75E+03	NA	NA
Predicted Stiffness	1.42E+05	1.93E+04	3.40E+03	1.11E+07	5.56E+06
Ratio Predicted/Measured	1.18	1.01	0.91	NA	NA

When I first collected the data and compared it with my spreadsheet, my roll stiffness agreed reasonably well with my collected data. However, my measured yaw stiffness was approximately 1/10th the predicted yaw stiffness and my pitch stiffness was approximately 1/20th the predicted yaw stiffness.

Originally, my model for both the yaw and pitch stiffness treated the entire carriage as a simply supported beam with various moment loads being applied. I played with the sensitive parameters in the analysis and determined that no reasonable value of the parameters of the carriage would yield the stiffness of my actual carriage. After re-analyzing my carriage, I determined that since the top of the carriage consisted of 25.4mm thick aluminum, any deflection measured was most likely due to the side plate and bottom plates of the carriage bending. I updated my analysis to reflect that the pitch stiffness was equal to a torsional load applied on the bottom plates of the carriage and the yaw stiffness was a torsional load applied to the side plates of the carriage. With only a few parameter tweaks (thickness, length, etc,) my predicted stiffness matched reasonably well with what I was actually measuring.

The yaw and pitch accuracy was determined by finding the distance between the average values of the clusters of data points as the carriage traveled from the rear to the front of the rail. The vertical distance between the average values of the clusters is indicative of the pitch errors due to the flatness of the slide, similarly for the horizontal distance between the clusters and the yaw errors. The repeatability of this motion was found by finding the standard deviation of each of the clusters

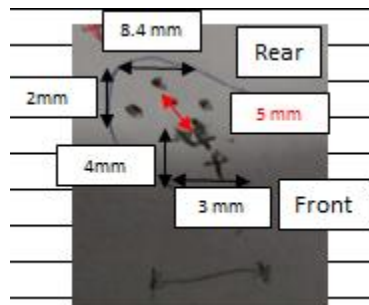


Figure 10. Flatness error data collection

Table 3. Yaw/Pitch Accuracy and Repeatability during travel along rail

Accuracy Yaw (rad, mrad)	0.000375	0.375
Accuracy Pitch (rad, mrad)	0.0005	0.5
Repeatability Yaw (rad, mrad)	0.000375	0.375
Repeatability Pitch (rad, mrad)	0.00025	0.25

In summary, there is actually quite a lot of error as the carriage travels the length of the rail. One reason for this is that the bearing pads used in testing were manufactured on the band saw and I believe that the testing reflects this manufacturing method. I have since corrected this by using bearing pads cut from a single sheet of Delrin plastic. The testing of the LMM using the properly manufactured bearing pads as well as with the addition of the lead screw can be seen in the Testing section of this report, section 6.

4. Design of the Spindle

I began my design of the spindle by developing a FRDPARRC as well as testing a sketch model concept.

SPINDLE FRDPARRC					
Functional Requirement	Design Parameter	Analysis	References	Risks	Counter Measures
minimize geometric and load induced errors at part	>Shaft/Bearing Stiffness and Configuration	> Beam Bending and Deflections	>Roark's Formulas for Stress and Strain	leadscrew is too stiff and shears	>calculate bolted connection parameters
	>Tolerances of bearings, housing and shaft	>Abbe error due to tolerances		sloppy tolerances lead to high error	> find a way to make this extremely accurately
Must fit onto carriage	>Stiffness and Cost	>Beam Bending and Deflections	>Roark's Formulas for Stress and Strain	>Certain designs have varying stiffness depending on carriage	>Either design a constant stiffness LMS or design cutting range of travel to be at a certain point
Cost less than \$50	Ratio of 'Goodness' to \$\$\$	Cost analysis of bad proper bearings vs bushings	> McMaster-Carr, Igus, SDPSI, Misumi	> I am unable to design a proper lathe because I don't want to spend big bucks	>Make/buy super tight tolerance bushings
Driven By a motor/hand	Torque to spin when experiencing cutting forces	>friction and load calculations	>Shigley	>Cutting forces are stronger than predicted, motor stalls	> Thorough analysis of loads expected to be seen by carriage and frictional forces
	>System Friction		>Roark, FUNDaMental		
	>Cutting Forces	> Lathe Turning Force	> Machinery's Handbook		

Figure 11 FRDPARRC of the Spindle

I chose to make a sketch model consisting of nylon extrusion acting as bushings with a turned down aluminum shaft and an aluminum housing.

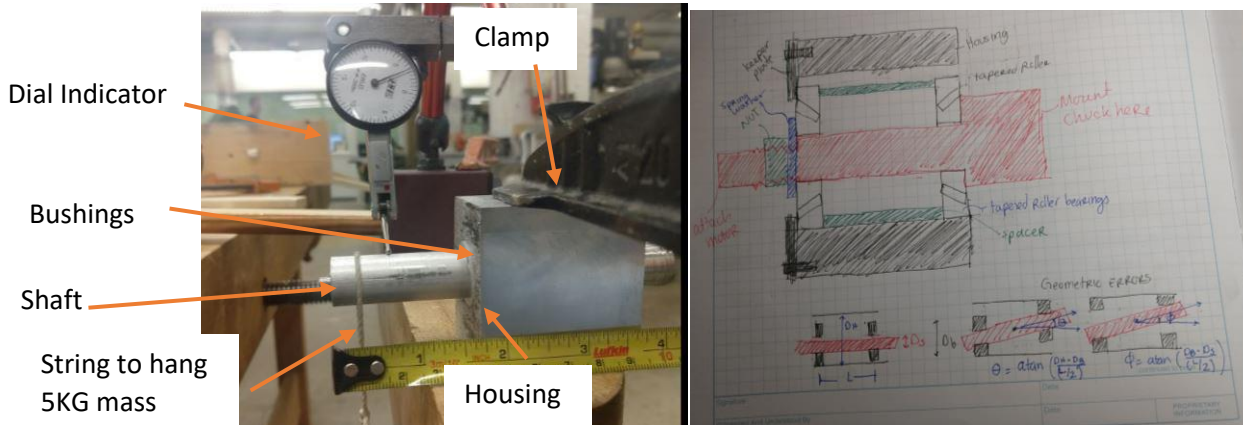


Figure 12 The Sketch Model of the Spindle and the sketch design

I was able to successfully measure both the geometric errors and the load induced errors, but my data from the load induced errors was lost when my computer crashed. These values were lost after the hobby shop was closed, so I was not able to reproduce the testing.

Table 4 Results from the testing of the spindle sketch model

Deflection From Loads (Y,X), mm, um Predicted	0.00575	5.76
Deflection From Geometry (Y,X) , mm, um Predicted	0.15278	152.78
Deflection From Loads (Y,X), mm, um Measured	NA	NA
Deflection From Geometry (Y,X) , mm, um Measured	0.1524	152.4

However, even using the model as a reference, geometric errors dominate the system. The model accurately predicted the geometric errors seen at the spindle to be 152 um, which is two orders of magnitude greater than the predicted deflection from loads. Therefore, ensuring tight tolerances on the diameters of the shaft and housing for the bearings will be crucial to ensuring minimal 'wobble' at the part/tool interface.

4.1. Design of the Rotary Motion Module

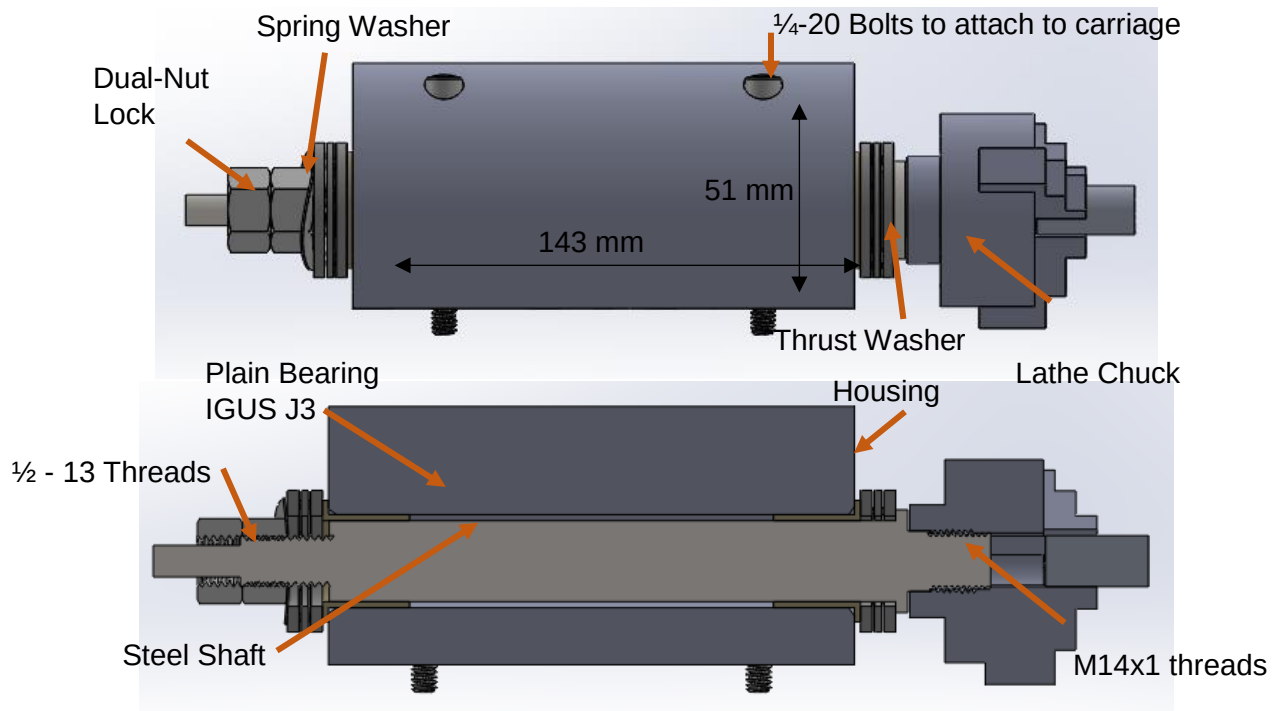


Figure 13 The basic design of the spindle unit

Knowing I would be pressed for time to complete the lathe, I tried to keep my spindle as simple as possible. I wanted to take advantage of the plain bearings that IGUS provided us with in our sample kits. I chose to use the J3 Iglide flanged bearing (<https://www.igus.com/product/81>) from the sample kit because it was labeled as resistant to unclean environments with longer working life than its counterparts.

To alleviate some of the friction due to axial loading, I chose to incorporate two thrust washers on either end of the spindle to interface between the flanged bearing and the spindle shaft. Two nuts on the end of the spindle tightened together will provide the spring washer with the preload required for the spindle. In my case, the thrust washers nor the plain washers require a preload to fulfill their function, therefore the preload of the spring washer should be chosen as to keep the spindle shaft in place as it experiences longitudinal forces.

There are several features in this design that lend themselves to locating the spindle shaft within the housing and the chuck on the shaft. The first feature is the flanges on each of the plain bearings. These bearings are designed to be press-fit into the housing and the flanges provide the axial reference location. A flange on the spindle shaft allows the front of the spindle to be pressed against the thrust bearing on the side with the shaft and it is fixed axially using the spring washers and dual-nuts. There is also small step on the inside of the chuck so that you are able to know exactly how deeply the spindle is threaded onto the chuck.

The key features requiring tight tolerances are the diameter of the shaft that fits within the J3 bearings, the diameter of the shaft that sits upon the inside of the chuck and the bore diameter of the housing.

4.2. Stiffness

In order to properly determine the load induced deflections within the spindle, it is necessary to develop a model for the stiffness of each component when experiencing radial and axial loads. A summary of the model used can be seen below in Fig. 2.

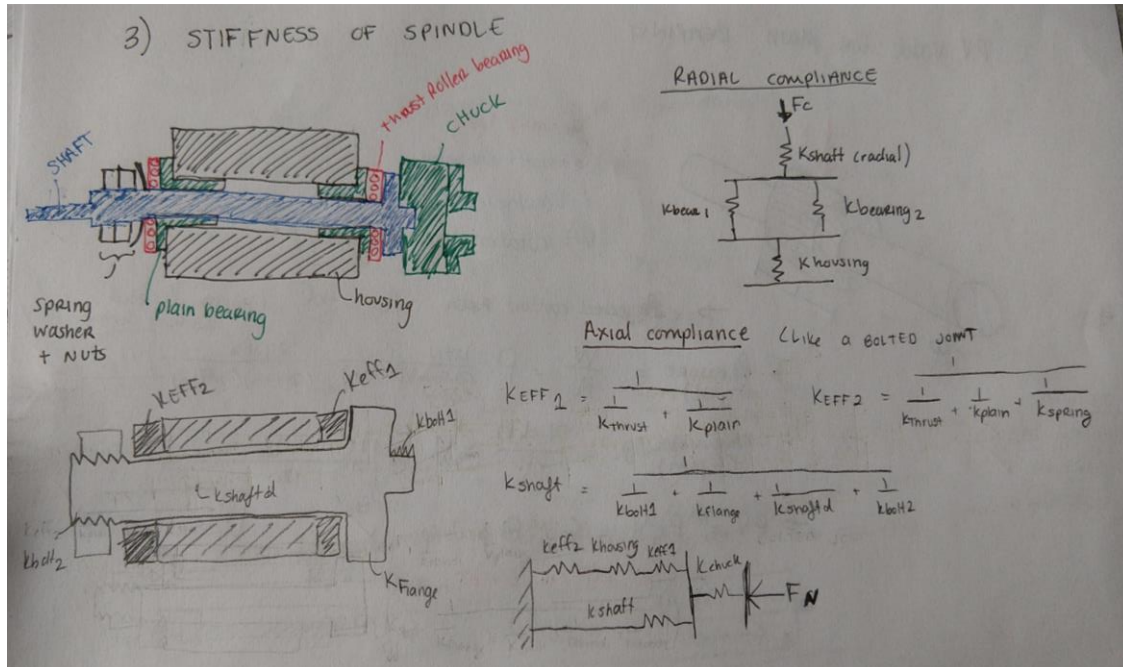


Figure 14. The radial and axial stiffness models for the spindle

In short, to model the radial compliance of the spindle at the center of stiffness, it is necessary to compute the effective stiffness of the members carrying the radial loads, in this case the cutting force. This can be done by putting the stiffness of the shaft in series with the bearing and housing stiffness. Similarly, to model the axial stiffness, it can be thought of as the shaft acting as a bolt that is bolting the plain bearings, thrust washers and housing together.

The full derivation for the individual stiffness for each component can be seen at the end of this writeup. Please see LatheDesign_RMH.xlsx for the full calculation.

One major concern that may be my downfall is the generation of frictional heat during the spinning process. To ensure that the temperature rises will not cause the spindle to jam due to thermal expansion, I estimated the thermal expansion of the shaft and the J3 bearings by finding the power dissipated due to friction. It is also necessary to calculate the PV value of the plain bearings to ensure that the applied loads due to cutting will not overstress the bearings.

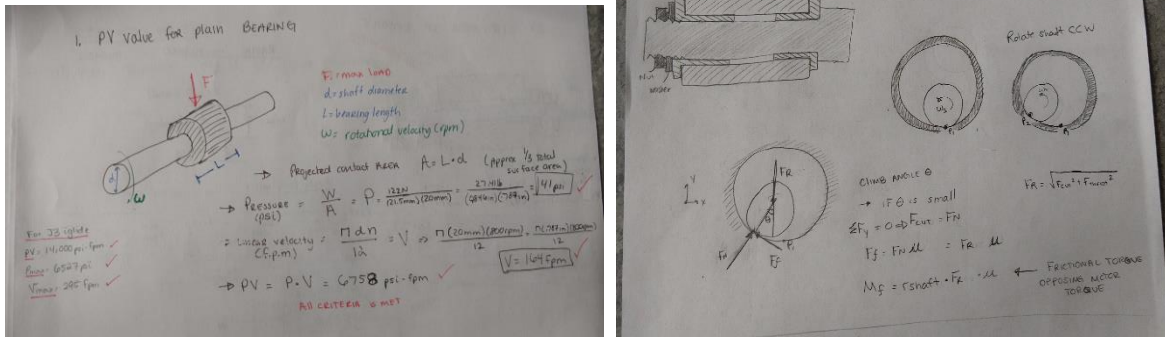


Figure 16. PV calculation and Frictional torque calculation

By calculating the power dissipated due to friction, I believe a rough estimation of the thermal expansion induced due to frictional heat can be calculated as follows. By knowing the allowable expansion between the bearing and the shaft, we can calculate the temperature rise to induce that expansion. Then, we can use the power dissipated due to friction to find the time it will take to raise the temperature of the material to the required amount.

$$P_{diss} * time = mc\Delta T \quad (1)$$

$$\Delta r = r\alpha\Delta T \quad (2)$$

Where c is the specific heat of the material, and α is the coefficient of thermal expansion. Plugging (1) into (2) and solving for the time to heat yields,

$$time = \frac{mc}{P_{diss}} \frac{\Delta r}{r\alpha} \quad (3)$$

Although the coefficient of thermal expansion for the bearings was available from the data sheet, the specific heat capacity was not. In my calculation, I therefore estimated that it would most likely be similar to that of Delrin.

This calculation reveals that it will take approximately 1 hour for the steel shaft to expand enough to cause the bearings to jam. However, and more importantly, it will only take 18 seconds of cutting for the bearings to expand! Not a comforting result and a highlight of why plain bearings are not typically used in this type of application.

However, I believe I would like to proceed but I will ensure to add lubricant to decrease the coefficient of friction and add ventilation holes in my spindle to help dissipate heat.

5. Design of the tool holder and leadscrew actuation

5.1. Design of the tool holder

The tool holder was modeled after the tool holders found on standard machine shop lathes. Four $\frac{1}{4}$ -20 screws provide the clamping force needed to hold the tool in place during the cutting process. A $\frac{1}{2}$ -13 bolt hole threaded into the carriage allows the tool holder to be clamped to the carriage as well as allow for angle adjustment to the tool.

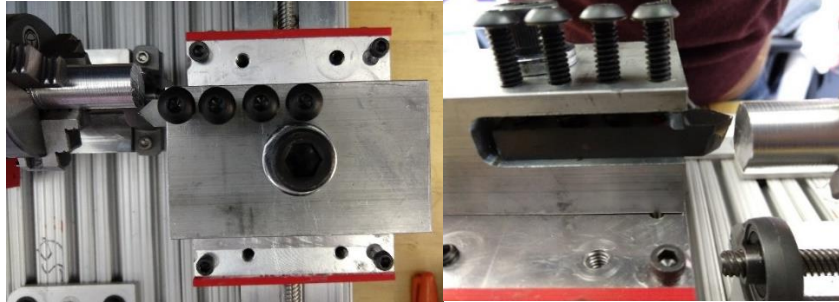


Figure 17 The Tool holder for the cutting tool

5.1. Design of the lead screw actuation

In addition to rotating the part that is to be cut, it is necessary to actuate the carriages. My chosen method to do so is to use a leadscrew attached to the center of the carriage. By design as well as by material availability, my carriage and rail design is extremely chunky and stiff. In fact, the translational Z axis and Y axis stiffness was not measurable using a dial indicator. Therefore, I determined that I needed either the connection between the carriage and the leadnut to be compliant, or the leadscrew mount to be compliant so as not to fight against the extremely stiff carriage.

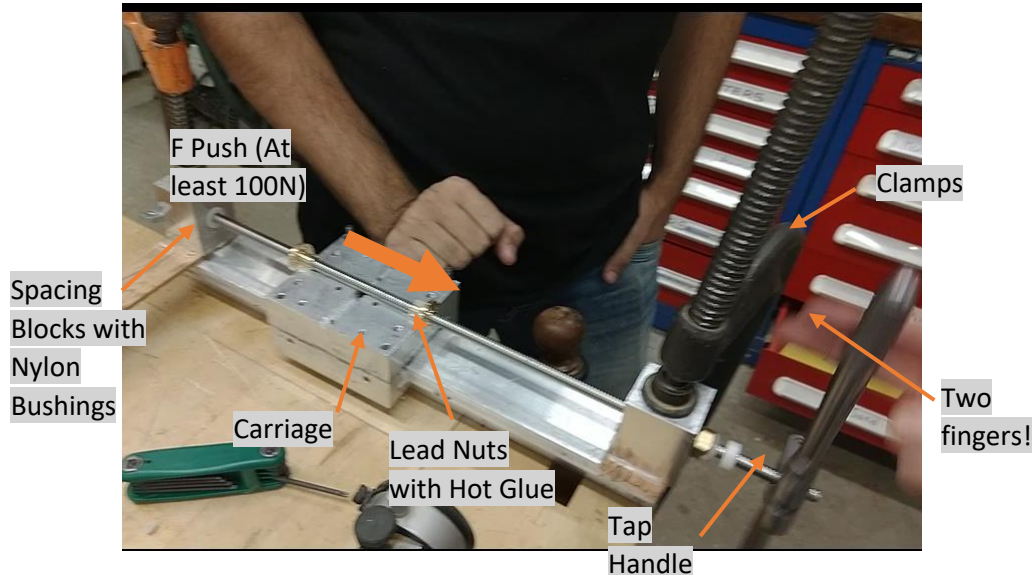


Figure 18. Sketch Model testing the connection between the Leadnut and the carriage

This part of the sketch model testing was more qualitative than quantitative. I wanted to see how the carriage would move using the cheap lead screw bought from amazon mounted in an intentionally non-accurate way. I wanted to ensure that even though I may mount the screw incorrectly, the forces would not cause undo stress to the carriage.

I tested the motion with applied loads by first having a friend push pretty hard (at least 100N) on a bolt sticking out of the side of the carriage (used for testing carriage stiffness). I was still able to move the carriage over its full range of motion (350 mm) without anything breaking and only using two fingers! This is very impressive to me considering the leadnuts were hot-glued to the carriage. I also hung a 10 Kg weight from the side bolt and was still able to move the carriage easily.

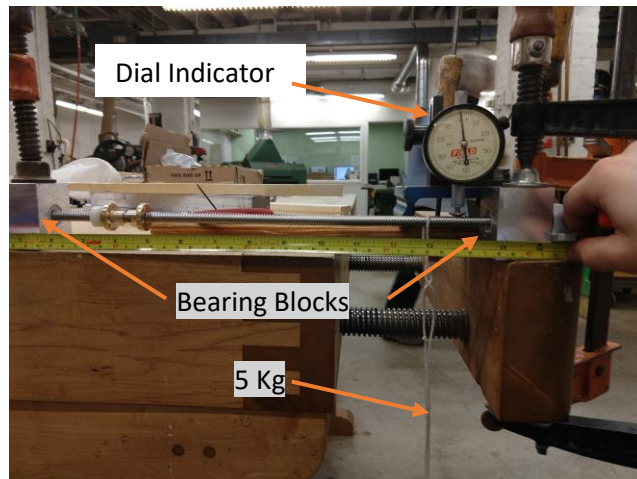
One concern that I had was the buckling of the lead screw. I considered four mounting cases for the lead screw and used the know parameters (mounting length, diameter, etc.) to estimate the buckling loads.

Buckling Calculation for Lead Screw			Youngs Modulus					
Lead Screw Length, mm	L	300	Steel	2E+11				
Screw Root diameter, mm	D	8	Aluminum	7E+10				
Screw Youngs Modulus, Pa	E	7E+10						
Moment of Inertia, m ⁴	I	2.0106E-10						
Buckling Load, N								
Configuration	Carriage@ L/2	Carriage@ L						
FI-FR	1543.42	385.86						
FI-SM	12625.20	3156.30						
SM-SM	6173.69	1543.42						
FI-FI	24694.78	6173.69						
Conclusion: No danger of buckling even when cutting at far edge of lead screw. Cutting Forces ~ 50-75 N								
Even with bad steel, the lead screw shouldn't buckle								

Figure 19. Predicted buckling loads of the lead screw

My conclusion based on these calculations is that even in the least stiff configuration (fixed- free) and assuming really poor quality steel (Youngs modulus= Aluminum), there is still a factor of safety of 5 when the cutting loads are considered.

I wanted to compare the stiffness model of the lead screw with measured deflection due to applied loads. This will be used when I estimate the deflection of the carriage due to forces from the lead/screw displacement.



Measured Deflection: Fixed-Fixed (mm)	0.16
Measured Deflection: Fixed-Simply Supp. (mm)	0.35
Predicted Deflection: Fixed-Fixed (mm)	0.12
Predicted Deflection: Fixed-Simply Supp. (mm)	0.2

Figure 20 Testing the stiffness of the lead screw

In this case, the measured values agree reasonably well with the sketch model and I believe the inaccuracies come from two sources, the first being the nylon bushings and the second being my use of a measuring tape and 'eyeballing' it. Experimenting in my spreadsheets shows that varying some of the parameters used by a few mm can bring the predicted values equal to what was measured.

The lead screws were mounted using the pillow blocks that came with the set (\$13 on Amazon). Although this is not a 'proper' mounting technique, it is sufficient for my application. If my lathe were to be sent to mass production, I would develop proper bearing supports and support blocks. I recommend that next year's class incorporate a second rotary motion module so that the students can build both a spindle and a lead screw. The model for the stiffness of my lead screw is based off of the work done by my classmate, Akshay Harlalka, for his rotary motion module. This stiffness is incorporated into my error budget spreadsheet (see Error_Budget_Spreadsheet_Rev2_RMH.xlsx).

Currently, there is no measure in place to eliminate the backlash from the leadscrew on the carriage. My initial cutting experiments lead that backlash elimination is not entirely necessary if the cutting occurs only from one direction.

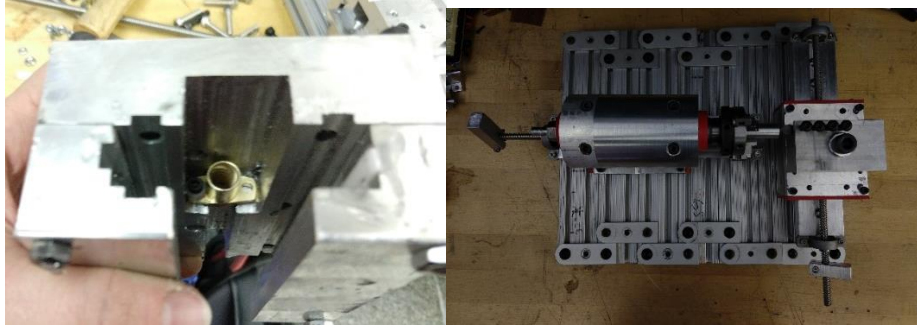


Figure 21. Leadscrew mounting to carriage and mounting to T-Rails

6. Testing the Lathe

The lathe was tested by performing a facing operation on test parts both with and without the spindle chuck. Facing passes were completed with the goal of removing 0.2 mm of material and create a perfectly flat piece (perpendicular to the profile of the part to be cut).

A piece of flexible hosing and hose clamps were used to connect a power drill to the end of the spindle shaft to actuate it. A video of the cutting process can be seen as part of the supplemental material.

Two parts were cut. The first part was faced while mounted in the three-jaw chuck and the second part was mounted directly onto the spindle shaft. This was to determine the accuracy of the lathe itself and the components I designed and fabricated personally compared with the accuracy of the complete lathe with the chuck.



Figure 22 The parts faced using the chuck (left) and directly attached to the spindle (right)

The parts that were cut were then tested in two ways. The first was that they were mounted in a circular indexing mechanism and had a dial indicator dragged across the surface in 30 degree increments. The purpose of this test was to determine the general contour of the part as a general representation of the accuracy of the lathe. The second test was using the interferometer to determine the surface finish of the part as a representation of the repeatability of the lathe.

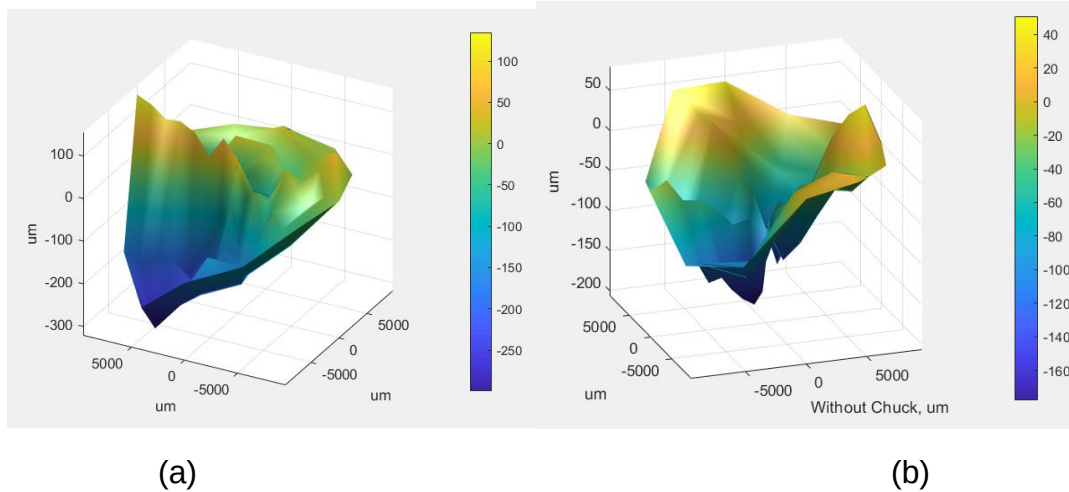


Figure 22. The contour plots of the test parts made (a) with the chuck and (b) attached directly to the spindle shaft

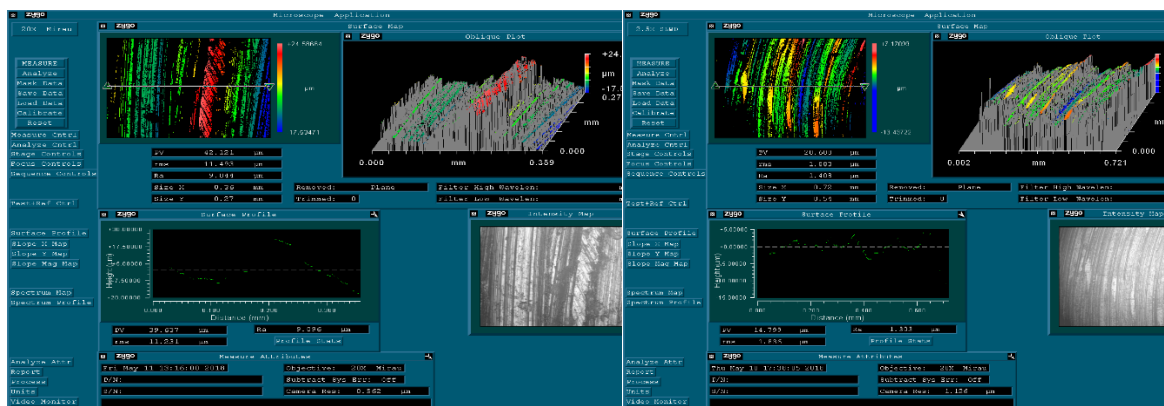


Figure 23. Surface finish plots taken using the interferometer of the test parts made (a) with the chuck and (b) attached directly to the spindle shaft

The contour plots made using data taken with a dial indicator suggest a very erratic shape. It appears that the shape of the test part made with the chuck has a large dip and peak on one side of the part. However, the overall shape is slanted across the entire face of the part. This could potentially be the effect of the tool coming loose during the cutting process

The contour plot from the part made without the spindle shaft suggests that the two linear axes are not exactly perpendicular. The shape suggests that the tool tip is angled as depicted in the following figure.

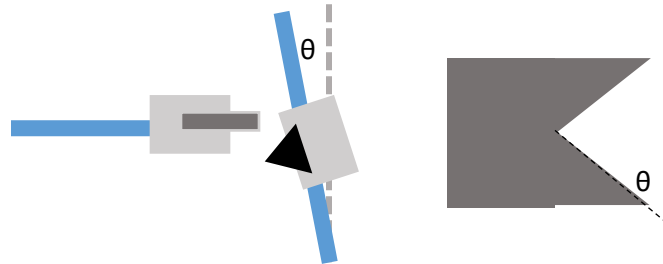


Figure 24. Perpendicularity error of the t-slides that created the observed contour in the test part

Using the contour plot, this angle was calculated to be 1.45 Degrees. This angle, if corrected should increase the accuracy of the manufactured part quite significantly.

I took several sets of data using a 20X and 2.7X magnification on both test parts. The average values obtained can be seen in the table below.

Table 4. Surface finish parameters from the test parts

Parameter	Value, μm
Part made with chuck	
Average PV (2.5X)	78.2
Average RA (2.5X)	10.6
Average PV (20X)	37
Average RA (20X)	5.75
Part made without chuck	
Average PV (2.5X)	52.37
Average RA (2.5X)	5.23
Average PV (20X)	20.46
Average RA (20X)	2.52

These values suggest that the lathe has a repeatability approximately equal to the surface finish RA value of 2.52 μm without the chuck and 5.75 with the chuck.

The contour plot suggests the part shape has a standard deviation of 56.23 μm with the chuck and 93.95 μm with the chuck. If we consider 2 standard deviations for a 95% confidence interval, I believe this means that a part faced without the chuck will be flat (accurate) to within 112.46 μm +/- 2.52 μm and a part faced with the chuck will be accurate to within 187.9 μm +/- 5.75 μm .

I believe that the contour plot determines the overall shape accuracy (flatness) of the part while the surface finish is indicative of repeatability of the lathe.

I believe this means that the overall result is just outside original goal of accuracy to within 100um. However, my error budget spreadsheet predicts an error of up to 365 um. I believe the discrepancy comes from slight errors in manufacturing the actual system, but overall I am satisfied with the spreadsheet's ability to predict the errors in the part.

All Axes	For the Entire Machine				
	Number of axes	All axes' Geometric Errors			F=kX displacement
	7	Random		Systematic	
		Sum	RSS	Avg(SUM, RSS)	Sum
	deltaX	0.342965	0.185802	0.264384	0.000000
	deltaY	0.276059	0.157203	0.216631	0.000000
	deltaZ	0.165911	0.091597	0.128754	0.000000
	Vector displacement	0.470489	0.260048	0.365247	0.000000

Figure 25. The error budget spreadsheet results

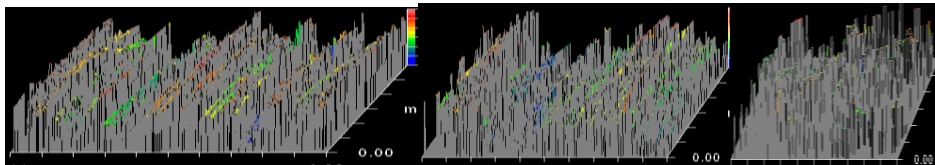


Figure 26. Surface roughness of the part made without the chuck in 2mm increments